

Model Question Paper
FIFTH SEMESTER DIPLOMA IN WOOD AND PAPER TECHNOLOGY
STATISTICAL QUALITY ASSURANCE

PART.A

Time 3Hrs

Maximum marks 100

I. Answer the following questions in one or two sentences (2x5=10)

1. State quality audit.
2. Compare attributes and variance
3. Mention significance of LTPD in OC curve
4. How measure of central tendency is calculated
5. State process capability.

PART .B

II. Answer any five questions among the following (6x5=30)

1. Discuss the step by step procedure of quality control programme.
2. Calculate measure of central tendency and dispersion of the following data obtained during an inspection 4,7,6,9,8,5,6
3. State and explain quality characteristics of a commodity.
4. Classify control chart and list out its objectives
5. Compare inspection and quality control.
6. Explain double sampling plan by taking suitable example
7. A certain type of electric components has a uniform failure rate of 0.00001 per hour. What is its reliability for the specified period of service of 10000 hrs

PART.C

Answer one full question from each module (15x4=60)

Module.1

- III. (a). State and discuss quality material management (7)
(b). Explain objectives of quality control (8)

OR

- IV. (a). Explain the different stages of quality assurance programme (8)
(b). Discuss vendor-supplier roll in quality assurance (7)

Module.II

V. (a). Construct a frequency polygon for the measured value given below

Class value	20-24	25-29	30-34	35-39	40-44	45-49	50-54	54-59
frequency	1	2	3	6	5	4	4	2

(8)

(b). Discuss process capability of a production process with the help of normal distribution curve (7)

OR

- VI. (a). Calculate dispersion of following measurements 15, 12, 5, 7, 9, 5, 11, 28 (7)
(b). Test of the stiffness of a number of aluminum alloy channel given below calculate mean and standard deviation (8)

Stiffness	195	192	189	186	183	180	177	174
Frequency	1	2	7	11	25	13	6	5

Modele.III

VII. (a) Ann inspection of 10 samples of size 400 each revealed the following number of defective items. With help of np chart comment on process capability of production process

17, 25, 14, 26, 9, 4, 19, 12, 9, 15 (9)

(b).Explain step by procedure of setting of control chart (6)

OR

VIII. (a).Construct a range chart for following observations and comment on process capability.

Assume $D_3=0.14$ and $D_4=1.86$ Ranges-3, 4, 3, 3, 4, 5, 4, 3, 4, 3 (7)

(b).Comment on process capability by constructing a mean chart for the following observations

Sample mean	42	45	41	44	40	45	46	44
Sample range	3	2	4	5	1	3	2	4

(8)

Module.IV

IX. (a).State and explain indices of acceptance sampling plan (7)

(b).Compare Random sampling plan and stratified sampling plan (8)

OR

X. (a) .Discuss Significance of OC curve for the study of single sampling plan (8)

(b. An element has a probability of successful operation over a given period of 60 percent. If four such elements are connected in parallel estimate the improvement factor.

Given $p_1=p_2=p_3=p_4=0.6$ (7)